Sai Sandeep Kumar Vijayarao

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RESEARCH INTERESTS

Large Language Models · Agentic AI Systems · Multimodal Reasoning · Neural Routing Architectures · On-Device Machine Learning · Privacy-Preserving AI · Foundation Model Fine-Tuning · Human-AI Interaction · Responsible AI

EDUCATION

Northeastern University *Sep 2025 – May 2027*
Master of Science, Computer Science *San Jose, CA*

- Specializations: Natural Language Processing, Multimodal AI, AI Agents, Large Language Models, Computer Vision, Responsible AI

R.M.K. Engineering College (Anna University) *Oct 2021 – May 2025*
Bachelor of Engineering, Computer Science with Honors in Artificial Intelligence *Chennai, India*

- IEEE AISP 2024 publication: real-time human pose estimation achieving **85% mAP** and **0.78 IoU** via MediaPipe + custom keypoint regression

PROFESSIONAL EXPERIENCE

Northeastern University *Sep 2025 – Present*
Graduate Teaching Assistant — Computer Vision, Generative AI, HCI *San Jose, CA*

- Delivered instruction and technical support across **3 graduate courses** (Computer Vision, GenAI, HCI) serving **100+ MS students**, providing weekly office hours and graded assessments for complex ML pipelines.
- Developed curriculum modules on transformer architectures, diffusion models, and vision-language grounding used by **30 students per course**.

Trominosoft *Feb 2025 – Aug 2025*
AI Developer *Hyderabad, India (Remote)*

- Reduced inference latency by **30%** across a production LLM serving pipeline handling **10K+ daily requests** by implementing request batching, KV-cache tuning, and async routing.
- Cut orchestration overhead by **40%** and hallucination rate by **38%** by redesigning the multi-agent prompt pipeline with structured retrieval and output-validation layers.
- Improved RLHF alignment score by **19%** by fine-tuning a domain-specific LLM with DPO + LoRA on curated preference data.

Infosys Springboard *Sep 2024 – Dec 2024*
AI Engineering Intern *Remote*

- Reduced analyst task completion time by **50%** and achieved **3.2**× throughput improvement by deploying an LLM evaluation and fine-tuning framework with INT8/INT4 quantization via vLLM.
- Automated dataset curation and benchmarking pipelines for 4 model variants, enabling rapid A/B comparison of BERT, GPT, and LLaMA checkpoints.

Global Techno Solutions *Jan 2024 – Apr 2024*
Software Engineering Intern *Chennai, India*

- Improved hospital management workflow efficiency by **30%** by developing a React 18 + Node.js + MongoDB full-stack system with role-based access control for 5 clinical departments.
- Reduced page load time by **45%** through server-side caching and query optimization on a **50K+ record** MongoDB instance.

RESEARCH & PUBLICATIONS

IEEE AISP 2024 — Real-Time Human Pose Estimation via MediaPipe and Keypoint Regression *2024*

- Achieved **85% mAP** and **0.78 IoU** on multi-person pose estimation; deployed as a real-time inference system at **30+ FPS** on standard CPU hardware.
- Proposed a lightweight keypoint regression head reducing parameter count by **22%** while maintaining accuracy

within 2% of full-model baselines.

PROJECTS

ANU — Apple Swift Student Challenge 2026 Winner *2026*
Swift, CoreML, Apple Intelligence APIs, SwiftUI, on-device LLM orchestration

- Built a Swift-native iOS app with a **10-agent orchestration system** powered by Apple Intelligence APIs, winning the **Apple Swift Student Challenge 2026** as sole developer.
- Reduced frontier API token usage by **50%** via on-device CoreML routing that resolved the majority of user intents locally before escalating to cloud models.

Dual LLM Personalization System — Multimodal Routing (Google Omni Parallel) *2025*
Python, LangChain, OpenCLIP, ChromaDB, LLM routing, multimodal inference

- Achieved **92.6% routing accuracy** and reduced LLM cascade rate from **30.8% to 14.7%** by engineering a dual-model router that dispatches queries to lightweight or frontier LLMs based on complexity classification.
- Served **85.3%** of all queries at **$0 cost** using on-device lightweight inference; multimodal routing architecture was independently recognized by DeepMind researchers at Google I/O 2026 as parallel to Google Omni.

AIFORG Agentic Orchestration Framework — (Google Spark Parallel) *2025*
Python, Google ADK, LangChain, tool-use agents, multi-agent orchestration

- Reduced third-party agentic tool integration time to **under 2 hours** by building a modular orchestration framework with standardized agent interfaces, dynamic tool registration, and Plan→Reason→Act execution loops.
- Framework architecture was independently recognized by DeepMind AI researchers at Google I/O 2026 as structurally parallel to Google Spark, validating the agentic orchestration design.

On-Device Agentic System *2025*
Gemma 4 4B, CoreML, frontier LLM escalation, Swift, Python

- Reduced frontier API token costs by **50%** by deploying Gemma 4 4B as a host model on-device, routing only high-complexity tasks to cloud frontier models via a confidence-gated escalation policy.
- Implemented a structured escalation protocol achieving sub-200ms local inference latency for 70%+ of user intents.

Agentic Research Assistant *2025*
Google ADK, ChromaDB, RAG, Plan–Reason–Act, dual-layer memory

- Built a multi-agent research assistant using Google ADK with a **Plan→Reason→Act** loop and **dual-layer ChromaDB memory** (episodic + semantic) enabling context retention across multi-session research workflows.
- Reduced redundant retrieval calls by **35%** through hierarchical memory indexing that deduplicates episodic events against long-term semantic store.

LLM Benchmarking & Fine-Tuning Framework *2025*
DPO, LoRA, INT8/INT4 quantization, vLLM, HuggingFace, RLHF

- Improved model alignment by **19%** and achieved **3.2**× throughput over baseline by fine-tuning with DPO + LoRA on curated preference datasets and serving via vLLM with INT8/INT4 quantization.
- Built an automated evaluation harness benchmarking 4+ LLM variants across accuracy, latency, and cost, accelerating model selection cycles.

TryOn — 360° Video to 3D Human Reconstruction *2025*
Python, NeRF, 3D Gaussian Splatting, MediaPipe, PyTorch, OpenCV

- Achieved **85% reconstruction accuracy** vs. professional 3D cameras with **1–2 minute** end-to-end processing by building a NeRF + 3D Gaussian Splatting pipeline that converts 360° video into textured human meshes.
- Reduced per-frame pose estimation error by **18%** by integrating MediaPipe landmark preprocessing as input normalization to the volumetric rendering stage.

Real-Time Lane Detection *2024*
Python, YOLOv8, OpenCV, TensorRT, PyTorch

- Delivered **99% detection accuracy** at **150 FPS** on embedded GPU hardware by training a YOLOv8-based lane segmentation model with TensorRT INT8 optimization.
- Outperformed prior baseline by **80%** in inference speed while maintaining accuracy parity, enabling real-time deployment on autonomous driving edge hardware.

Vision-Language Grounding *2024*

Python, CLIP, Stable Diffusion, zero-shot classification, HuggingFace

- Built a zero-shot vision-language grounding system using OpenCLIP embeddings achieving **top-3 accuracy of 88%** across 12 open-vocabulary categories without any category-specific fine-tuning.
- Integrated Stable Diffusion as a visual augmentation backend, generating synthetic training views that improved cross-domain generalization by **14%**.

AI Agentic Second Brain *2024*
Python, LangChain, DigitalOcean, vector DB, long-term memory agents

- Deployed a persistent multi-agent knowledge system on DigitalOcean with long-term semantic memory, enabling coherent cross-session recall and document synthesis across **500+ stored knowledge nodes**.
- Reduced manual note retrieval time by **60%** through automated entity extraction, cross-linking, and relevance-ranked vector search.

SkillGap AI — 3× PMI San Francisco Award Winner *2024*
Python, React, LangChain, RAG, PostgreSQL, Vercel, Supabase

- Built and deployed an AI-powered skills-gap analysis platform serving **500+ concurrent users**, winning **3 PMI San Francisco Awards** as sole developer.
- Reduced career roadmap generation time by **70%** via a RAG pipeline that maps user profiles against live job market embeddings to surface targeted upskilling recommendations.

Neural Studio — Multimodal AI Canvas *2024*
Gemini API, React, multimodal generation, real-time canvas

- Built a Gemini-powered multimodal creative canvas at the Gemini Hackathon that processes text, image, and audio inputs in a unified workspace, enabling combined-modality generation workflows.

Omni-Axon *2024*
Python, multi-agent, Stanford × DeepMind Hackathon

- Designed and implemented a multi-agent coordination system at the Stanford × DeepMind Hackathon, focusing on dynamic task decomposition and inter-agent communication for complex reasoning pipelines.

Notification Urgency Prediction *2024*
Python, BERT, scikit-learn, NLP, 3-class text classification

- Trained a BERT-based 3-class text classifier to predict notification urgency (low / medium / high), achieving **91% F1** on held-out test data across diverse notification corpora.

Hospital Management System *2024*
React 18, Node.js, MongoDB, REST API, role-based access control

- Improved clinical workflow efficiency by **30%** by building a full-stack hospital management system with RBAC covering patient, doctor, and admin portals across 5 departments.

Real-Time Human Pose Estimation (IEEE AISP 2024) *2024*
Python, MediaPipe, PyTorch, custom keypoint regression, OpenCV

- Published at IEEE AISP 2024; achieved **85% mAP** and **0.78 IoU** on multi-person pose estimation by proposing a lightweight keypoint regression head operating on MediaPipe landmark priors.

AWARDS & RECOGNITION

- **Apple Swift Student Challenge 2026 Winner** — Sole developer, ANU iOS app with 10-agent on-device orchestration system (Apple Intelligence APIs)
- **Google I/O 2026 — DeepMind Researcher Validation** — DeepMind product and AI researchers independently recognized Dual LLM Personalization System as parallel to Google Omni (multimodal routing) and AIFORG as parallel to Google Spark (agentic orchestration)
- **3× PMI San Francisco Award** — SkillGap AI; NextGen Leaders category; sole developer
- **IEEE AISP 2024 Publication** — Real-time pose estimation (85% mAP, 0.78 IoU); published at undergraduate level
- **Elected Head Boy** — Representing **600+ students**; demonstrated institutional leadership

LEADERSHIP & COMMUNITY

- **GDG Sunnyvale** — Core Team member; organized AI/ML developer events and workshops in the Bay Area

- **IEEE IAYPC** — Active contributor to IEEE Young Professionals chapter events and technical programs
- **Apple / Google / NVIDIA Developer Programs** — Active participant across all three developer ecosystems

CERTIFICATIONS

- Google Project Management Professional Certificate
- CodePath AI Engineering Program
- Linux Foundation DevOps & Kubernetes Certification

TECHNICAL SKILLS

Languages: Python, Swift, JavaScript, TypeScript, Java, C/C++, Kotlin, SQL, R

Frameworks & Libraries: PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace Transformers, LangChain, Google ADK, SwiftUI, React, Angular, Node.js, Django

LLM / AI Stack: BERT, GPT, LLaMA, Gemma 4, Gemini, YOLO, MediaPipe, OpenCLIP, Stable Diffusion, vLLM, LoRA, DPO, RLHF, INT8/INT4 quantization, CoreML, Apple Intelligence APIs

Agentic Systems: Multi-agent pipelines, RAG, LLM orchestration, tool use, agent memory, Plan–Reason–Act loops, ChromaDB, vector DBs

Infrastructure: Azure, GCP, DigitalOcean, Docker, Kubernetes, Vercel, Supabase, MongoDB, PostgreSQL, TensorRT